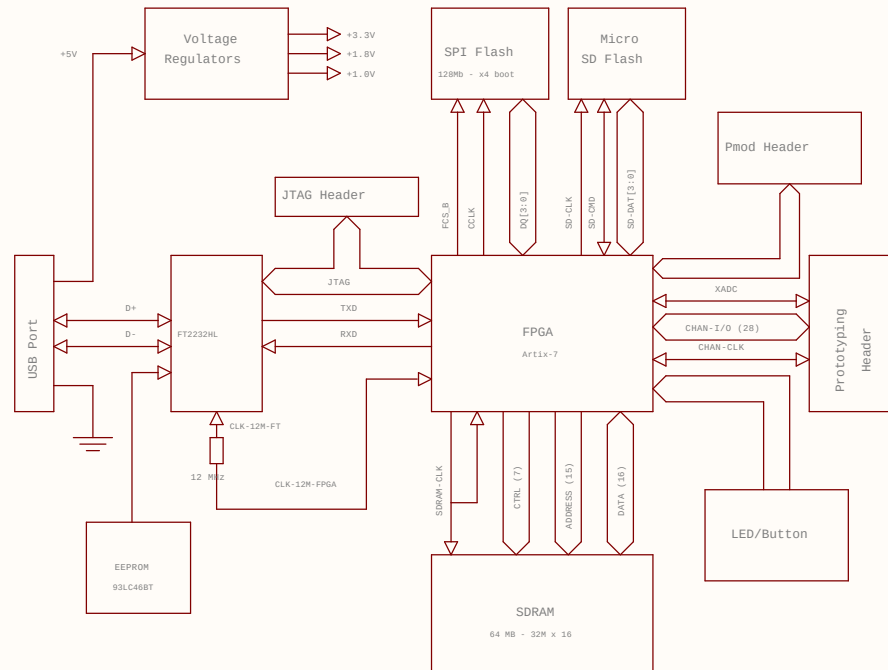


SPI flash sits on the FPGA's dedicated master-SPI config pins: CCLK K11, FCS_B K19, DQ[3:0] on D18/D19/G18/F18. Read at power-up.

microSD runs the native 4-bit SD bus, not SPI: CLK, CMD and DAT[3:0] on six bank-34 balls of its own, so a card cannot touch the config bus.

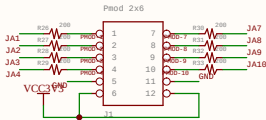
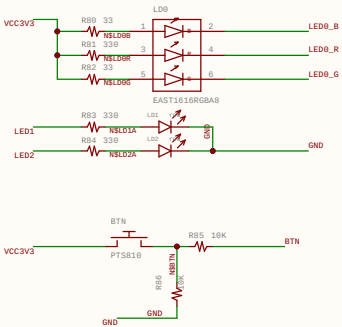


Block Diagram			
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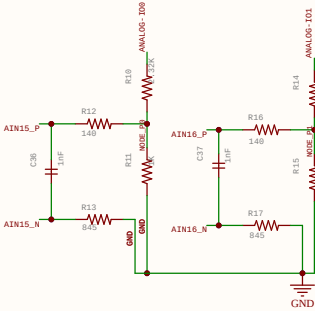


X2: 42 pins on a 2.54 mm grid in a 44-position pattern, rows 22.86 mm (0.900 in) apart.
Numbered right to left and straight through: top row 1-20, bottom row 21-44; positions 23 and 44 are empty since 2020-09-09.
TOP ROW, x 59.69 down to 1.27: GND, CHAN12, CHAN13, then CHAN-CLK and CHAN0..CHAN3 (pins 4-8), CHAN4 (pin 9), CHAN5..CHAN11 (10-16), +3.3V, +1.0V, +1.0V, GND.
FOUR positions between pin 9 and pin 10 -- x 36.83, 34.29, 31.75, 29.21 -- are RESERVED, not spare: the landing for the edge-mounted micro-USB X1, 7.80 mm of copper in 11.176 mm clear, centred with 1.69 mm each side.
BOTTOM ROW, x 59.69 down to 3.81: GND, VU, (23 empty), RST#, +3.3V, GND (pins 25-26), CHAN14..CHAN20 (pins 27-43), ANALOG-100, ANALOG-101; position 44 (x 1.27) is empty.

ZULU-CONN



Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the Pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmod A7 has none at its Pmod either. JA1-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16. JA7 is C15, also bank 16. JA8/JA9/JA10 are W16/W27/W27 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.



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General IO

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U3 = AS4C32M16SB-7TCN: 512Mbit (32M x 16) = 64MB.
Pin-for-pin and dimensionally identical to the 256MB
AS4C16M16SA-6TIN it replaces -- all 54 pins and every
TSP: 1 dimension checked against both datasheets, and
pin 40 is NC on both.
One DIFFERENCE, and IT IS NOT A PIN: the column address
is A0-A9 (1024 columns), not A0-A8 (512). The memory
controller must be set to 10 column bits, or it reaches
only half the array. Row A0-A12, banks BA0/BA1 and
refresh (8192 cycles / 64ms) are unchanged.
-7TCN since 2026-09-08 (-6TIN dry): 143 MHz max at CL13,
tRCD/trP 21, tRC 63 ns, TA 0-70 C, IOD1 max 110 mA.

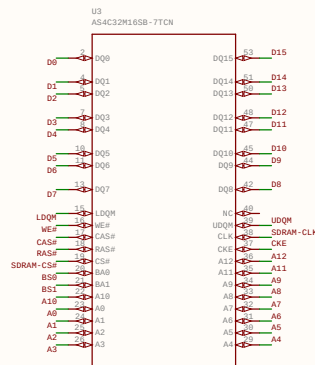


Figure 1: Schematic diagram of the proposed 8-bit SAR ADC. The diagram shows a ladder network of 8 bit cells. Each bit cell consists of a top branch with a PMOS transistor (labeled C3) and a bottom branch with an NMOS transistor (labeled C4). The gates of these transistors are connected to a common input signal VCC3V3. The sources of the NMOS transistors are connected to ground (GND). The drains of the NMOS transistors are connected to the gates of the PMOS transistors in the next stage. The output of the ADC is taken from the drain of the PMOS transistor in the final stage (labeled C9). The diagram also shows the connection of the input signal VCC3V3 to the gates of the PMOS transistors in each stage. The output of the ADC is labeled VDDQ08.

Memory			
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